

Amazon User Recommendations AWS Deployment

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Dataset Introduction: Amazon Transaction Data

product_id	Product ID
product_name	Name of the Product
category	Category of the Product
discounted_price	Discounted Price of the Product
rating	Rating of the Product
rating_count	Number of People Who Voted for that Amazon Rating
about_product	Description of the Product

user_id	ID of the user who wrote the product review
review_id	ID of the User Review
review_title	Short Review
review_content	Long Review
img_link	Image Link of the Product
product_link	Official Website Link of the Product

Data Cleaning & Preprocessing

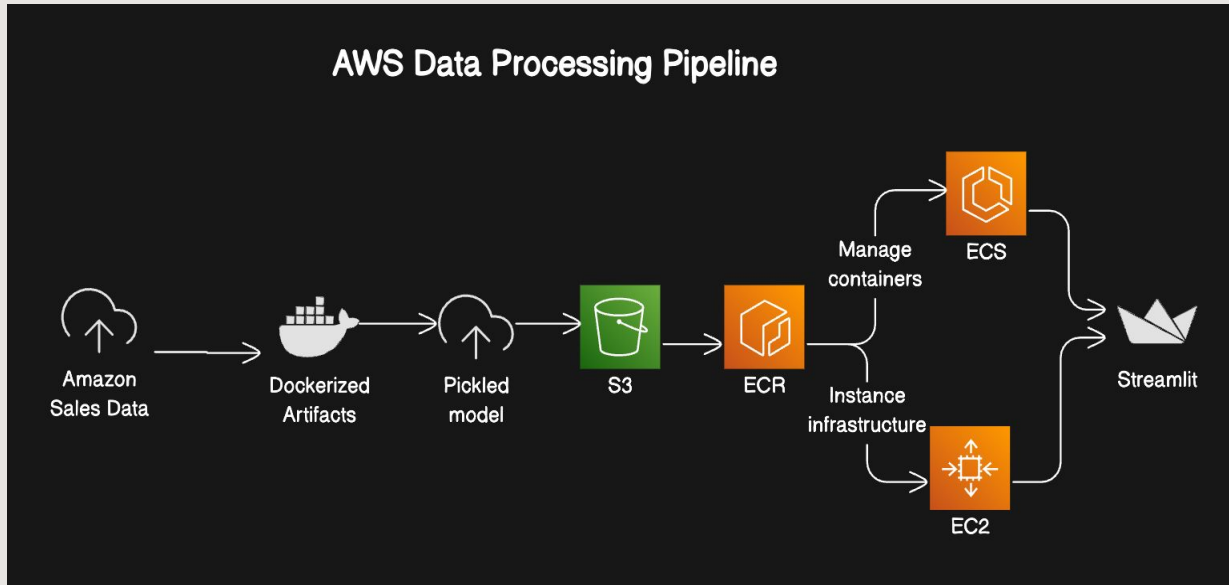


**Drop Null
Values**

**Added variable
that describes
length of
review content**

**One-Hot
Encoded
Product
Category**

Architecture Diagram



AWS Budget

Service	Cost (Monthly in USD)	Details
Amazon EC2	14.16	Number of instances: 1
S3 Standard	0.09	S3 Standard storage (1GB for feature and data transfer per month)
AWS Fargate	19.62	1 task per day, 20 GB ephemeral storage for ECS, 3 GB memory allocated
Amazon Elastic Container Registry	0.38	2 GB data inbound per month, 2 GB data stored per month

Model Building

- **Goal:** Predict user ratings and recommend the top 5 items to the user
- **Method:** Content Based & Collaborative Filtering Recommendation Engine
- **Collaborative Filtering:**
 - Features:
 - User_Id
 - Product_Id
 - Ratings
 - Models:
 - Grid_search SVD
 - Grid_search KNNBasic
 - Grid_search NMF
- **Content Based Filtering:**
 - Features:
 - Discounts
 - Actual_price
 - Category (one-hot)
 - Model: XGBoost Classifier
 - : L1, L2 regularization

Model Training Result

Models	SVD	KNNBasic	NMF
Best coefficients	n = 50, lr = 0.01, reg = 0.04	n = 40, similarity = cosine	n = 30, reg_u = 0.06, reg_i = 0.1
Test RMSE	0.15	0.266	0.29

- Finally, we got a test RMSE from the XGBRegressor for the ratings based on these features, as 0.24
- The model does not account for user-item interactions, so we could do future to take a weighted average of the ratings of XGB model and the SVD model, taking a 0.8:0.2 ratio in favour of the SVD model.
- Add run pipeline command in the docker file so that the model can be trained in EC2 instance and upload the artifacts to the S3 bucket.

Model Deployment

Dockerfile:

- Runs both model training and stream lit app deployment
- Allows for model training to utilize ECS compute

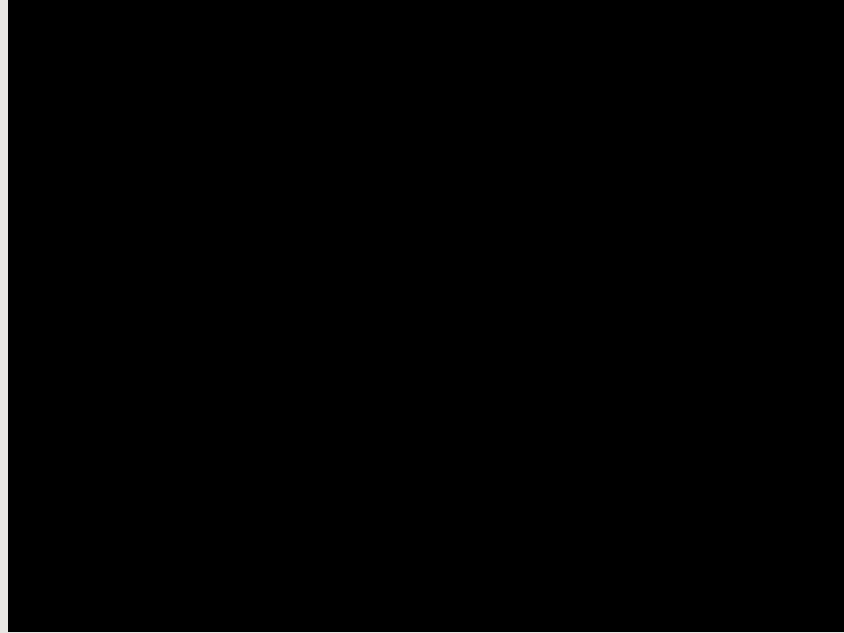
Docker image:

- Hosted in ECR
- Deploy in ECS instance with Application Load balancing to create an exposed web application

Data/Modeling Artifacts:

- All data and model artifacts are stored in S3 bucket

Demo



TEAM 9

<http://amazon-app-lb-732330289.us-east-2.elb.amazonaws.com/>

THANK YOU

For Listening